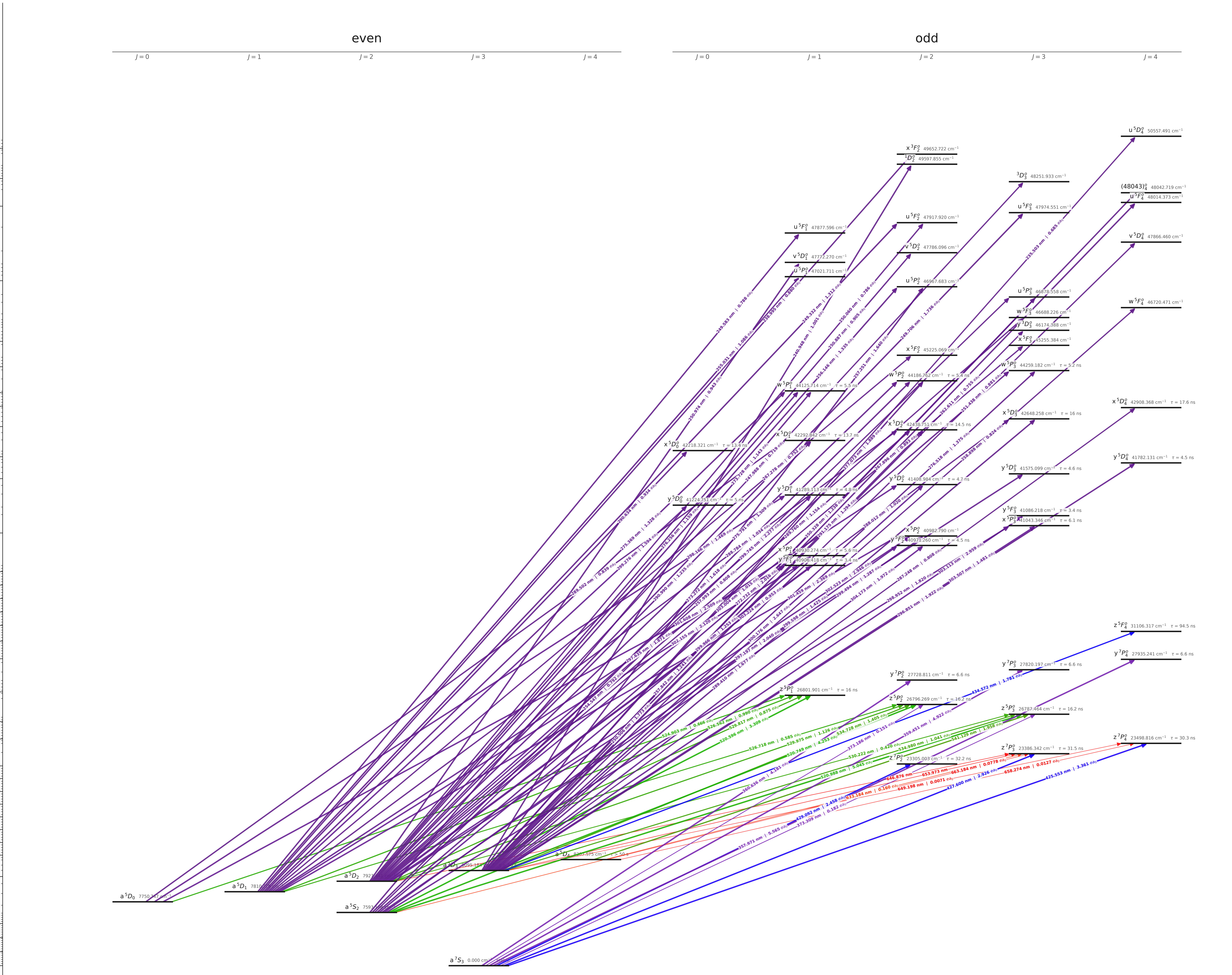


⁵²Cr (Cr I), nuclear spin $I = 0$. Strongest classified transitions below 2 μm (NIST): 56 levels, 87 lines.

Cr⁺ ionisation limit 54575.60 cm⁻¹ = 6.76651 eV (183.232 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| er \| J' \rangle |$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U(erJ)] (eao)	A (s ⁻¹)	F/2 π up. (MHz)	branch (%)	Use
$a^1S_3 - y^1P_2^o$	357.9708	357.8686	837.4775	5.565	1.520e+08	24.11	100	strong UV resonance line a753 - y7P4
$a^1S_3 - y^1P_3^o$	359.4511	359.3485	834.0285	4.922	1.510e+08	24.11	99.7	strong UV resonance line a753 - y7P3
$a^1S_3 - y^1P_2^o$	360.6357	360.5329	831.2888	4.181	1.510e+08	24.11	99.7	strong UV resonance line a753 - y7P2
$a^1S_3 - z^1P_2^o$	425.5534	425.4336	704.4768	3.361	3.300e+07	5.253	100	nm cooling line: Zeeman slower, MOT, imaging
$a^1S_3 - z^1P_3^o$	427.6000	427.4797	701.1049	2.926	3.170e+07	5.053	99.9	optical pumping to m _J = -3 and "depumper" towards 552 /...
$a^1S_3 - z^1P_2^o$	429.0924	428.9717	698.6664	2.458	3.100e+07	4.943	99.8	429.1 nm line
$a^1S_3 - z^1P_3^o$	633.1842	633.0091	473.4680	0.160	2.900e+04	5.053	0.0914	633.2 nm repumper 552 -> 7P3
$a^3D_3 - z^1P_3^o$	646.8765	646.6978	463.4462	-	-	5.053	-	646.9 nm repumper 502 -> 7P3
$a^3D_3 - z^1P_2^o$	649.1976	649.0182	461.7892	0.007	4.200e+01	5.253	0.000127	intercombination leak of the cooling line into metastable...
$a^3D_3 - z^1P_3^o$	653.9727	653.7921	458.4174	-	-	5.053	-	654.0 nm repumper 503 -> 7P3
$a^3D_4 - z^1P_3^o$	658.2741	658.0923	455.4219	0.013	1.270e+02	5.253	0.000385	intercombination leak of the cooling line into metastable...
$a^3D_4 - z^1P_2^o$	663.1842	663.0011	452.0500	0.078	6.000e+03	5.053	0.0189	663.2 nm repumper 504 -> 7P3

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$a^7S_3 - z^7P_0^o$	425.553	53-52	10.7(17)
$a^7S_3 - z^7P_3^o$	427.600	53-52	18(23)
$a^5S_2 - z^7P_3^o$	633.184	53-52	-155(53)
$a^5D_3 - z^7P_3^o$	653.973	53-52	670(28)
$a^5D_4 - z^7P_3^o$	663.184	53-52	819(28)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$